



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

CSDS 611, Advanced Topics and Methods of Data Science I Fall 2026

University of Tennessee, Knoxville

Course details: CRN 60635, credit hours 3

Meeting times: Tuesday and Thursday, 2:30pm–3:45pm

Classroom: Ayres Hall, room 112

Faculty contact information

Instructor: Ioannis Sgouralis

Office: Ayres Hall, room 323

Email: isgoural@utk.edu

Office hours

- *Standard office hours:* Starting from August 28, I will be hosting office hours every **Friday, 11:00am–1:00pm**. During office hours I will foster an open discussion format.
- *Alternative office hours:* In addition to the standard office hours, I will be available after each class or by **appointment**.

Course description: Selected topics in mathematical Data Science. Topics include: foundations of Artificial Intelligence, supervised Machine Learning, unsupervised Machine Learning, reinforcement learning, statistical learning, artificial neural networks, graph neural networks, Bayesian networks, physics informed models, linear and non-linear time series analysis, data assimilation and forecasting, parametric and nonparametric methods, generation and analysis of synthetic and field data, surrogate models and simulation, development and implementation of Monte Carlo and Markov chain Monte Carlo algorithms.

Recommended Background: CSDS 511 or CSDS 612 or any course in the sequences CSDS 513/514 or CSDS 515/516 (equivalent to Math 525/526).

Course objectives: This is an advanced topics course with a special focus on the principles and methods of Bayesian Optimization (BO). The course is designed for students seeking a comprehensive, method-oriented, understanding of BO as a tool for active learning and an integral component in modern data science and AI workflows. The course will combine rigorous mathematical foundations with specialized algorithmic procedures and practical computational implementation. Students completing this course should expect to gain both conceptual mastery and practical skills for applying and improving BO in advanced data settings.

Course information: We will introduce and study the core modeling and decision-making loop in BO, including probabilistic surrogate modeling and sequential, acquisition-driven experimentation, with attention to both theoretical and computational aspects. A rigorous treatment of Gaussian processes and Bayesian decision theory will be provided, along with practical implementation details and special considerations allowing for high-performance computing.

Course policy: When you can do so, I expect you to attend the classes and participate in the class discussions. I require attendance to the degree possible; however, I will **not** grade attendance. For those of you that will not be able to attend, course material such as lecture notes and assignments will be posted online. During class time, please, be considerate of the instructor and your classmates. Come to class on time and stay until class ends. Silence and put away cell phones or other electronic devices. Participate in open discussions without talking to your classmates at inappropriate times.

Course announcements: Throughout the semester, I will make every important announcement in class and electronically via **Canvas**. To ensure that you have received all the latest updates, I advise that you check your inbox regularly.

Textbook and supplemental resources: There are **no** required textbooks for this course. For *additional reading*, I recommend the following reference:

- *Pressé and Sgouralis, Data Modeling for the Sciences, Cambridge University Press*

on which the introductory lectures will be based.

Software: You are encouraged to use any software package you prefer; however, I will only provide pseudocode that is implemented in Matlab. You can access, download, and install Matlab to your computer from OIT's website <https://oit.utk.edu/research/research-software/matlab-simulink/>

Course notes: I will be posting lecture notes on **Canvas** after each lecture. To avoid conflicts, *do not distribute* any course related material or content (including lecture notes, homework assignments, and in-class projects) outside the class.

Course assignments

- **Homework assignments:** Throughout the semester, we will have 3 assignment sets with due dates roughly every 6-8 lectures. Assignments should be *typed* and handed in electronically in PDF format via Canvas. In preparing your reports, you can use any software you prefer.
- **Assignment due:** Assignment due dates depend on how much material we will cover in each lecture. For this reason, I will be posting specific due dates during the semester.
- **In-class project:** This course will conclude with an in-class project assignment. Each project will be completed during class time in groups of 2-3 students. Group reports should be *typed* and handed in electronically in PDF format like the homework assignments.
- **Teamwork:** In preparing your homework assignments, *I encourage you to discuss* the problems and seek the solutions *together* with your classmates. However, I expect each student to submit for grading their own solutions. In preparing the in-class project, I expect students to work in small groups and submit only a single report per group.

Course evaluation

- **Exams:** In this course, we will have **no** exams. Monday, December 7, 3:30pm–6:00pm is reserved for a final exam, *but we will not make use of it*.
- **Grading:** Each homework assignment will account for 20% and the in-class project will account for 40% of your coursework. We will have no extra credit on this course.

On assigning final grades, I will use the following tentative scale:

	A: 90-100%,	A-: 87-90%,
B+: 83-87,	B: 80-83%,	B-: 77-80%,
C+: 73-77%,	C: 70-73%,	C-: 67-70%,
D+: 63-67%,	D: 60-63%,	D-: 57-60%,
	F: 0-57%.	

How to be successful

- *Attend all lectures and course sessions.*
- *Stay current with course material and homework.*
- *Discuss homework problems and solutions with your classmates.*
- *Seek help and talk with me during office hours.*
- *Review class notes often.*

Key Campus Resources for Students:

- [Center for Career Development and Academic Exploration](#) (Career counseling and resources; Handshake job search system)
- [Course Catalogs](#) (Listing of academic programs, courses, and policies)
- [Hilltopics](#) (Campus and academic policies, procedures and standards of conduct)
- [OIT HelpDesk](#) (865) 974-9900
- [Schedule of Classes/Timetable](#)
- [Student Health Center](#) (visit the site for a list of services)
- [Academic Success Center](#) (Academic support resources)
- [Undergraduate Academic Advising](#) (Advising resources, course requirements, and major guides)
- [University Libraries](#) (Access to library resources, databases, course reserves, and services)

Tentative course outline

- Meeting 1: Introduction to CSDS 611
- Meeting 2: Introduction to Bayesian methods
- Meeting 3: Bayesian inference
- Meeting 4: Gaussian distributions
- Meetings 5-6: Non-Gaussian distributions

1st homework assignment

- Meeting 7: Bayesian hypothesis testing and Bayesian summaries
- Meeting 8: Decision theory
- Meeting 9: Bayesian linear regression
- Meeting 10: Bayesian model selection
- Meeting 11: The kernel trick
- Meeting 12: Bayesian logistic regression and Laplace approximations

2nd homework assignment

- Meeting 13: Gaussian process regression
- Meeting 14: Covariance kernels
- Meetings 15-16: Bayesian optimization
- Meeting 17: Bayesian quadrature
- Meetings 18-19: Sequential decision theory and active searches

3rd homework assignment

- Meeting 20: Importance sampling and Markov chain Monte Carlo methods
- Meeting 21: Advanced sampling techniques
- Meetings 22-23: Implementation of Gaussian processes
- Meetings 24-27: In-class project
- Meeting 28: Catch-up

The instructor reserves the right to revise, alter or amend this syllabus as necessary. Students will be notified in writing / email of any such changes.

The Campus Syllabus

ACADEMIC INTEGRITY

Each student is responsible for their personal integrity in academic life and for adhering to UT's Honor Statement. The Honor Statement reads: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."

YOUR ROLE IN IMPROVING THE COURSE THROUGH ASSESSMENT

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester you may be requested to assess aspects of this course, either during class or at the completion of the class, and through the TNVoice course evaluation. Please take the few moments needed to respond to these requests as they are used by instructors, department heads, deans and others to improve the quality of your UT learning experience.

STUDENTS WITH DISABILITIES – <http://sds.utk.edu>

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kind of disability, or a temporary injury, you are encouraged to contact [Student Disability Services](#) (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter. You may also contact your SDS Coordinator if you have any questions, concerns, or experience any barriers to using your accommodations in this course.

ACCESSIBILITY POLICY AND TRAINING – <http://accessibility.utk.edu>

The University of Tennessee, Knoxville, provides reasonable accommodations for individual students with disabilities through its office of [Student Disability Services](#). The university is also committed to making information and materials accessible, when possible. Resources and assistance to support these efforts can be found at <http://accessibility.utk.edu/>.

WELLNESS – <http://wellness.utk.edu/> and <http://counselingcenter.utk.edu/>

The [Center for Health Education and Wellness](#) empowers all Volunteers to thrive by cultivating personal and community well-being. The Center can answer questions about general wellness, substance use, sexual health, healthy relationships, and sexual assault prevention. The [Student Counseling Center](#) is the university's primary facility for personal counseling, psychotherapy, and psychological outreach and consultation services.

[The Student Health Center](#) has a wide range of clinics and services. Their goal is to provide quality healthcare and assist you in maintaining a healthy lifestyle to support your success as a student. To schedule an appointment, call 865-974-5080. To report a health-related absence visit: <https://studentlife.utk.edu/caring-for-all-vols/absence/>.

Any student who has difficulty affording hygiene products, groceries, or accessing sufficient food to eat every day is urged to contact [Basic Needs](#) for support. The Big Orange Pantry, located on the 3rd floor of

the Student Union, is a free resource for all students, faculty, and staff, no matter how great or small their need is. Students who need emergency financial assistance can also request funding from the [Student Emergency Fund](#).

Students who are experiencing non-academic difficulty or distress and need assistance should call 974-HELP or [submit an online referral](#). The 974-HELP team specializes in aligning resources and support to students experiencing mental health distress.

EMERGENCY ALERT SYSTEM – <http://safety.utk.edu/>

The University of Tennessee, Knoxville, is committed to providing a safe environment for learning and working. When you are alerted to an emergency, please take appropriate action. Learn more about what to do in an emergency and sign up for UT Alerts. Check the emergency posters near exits and elevators for building specific information. In the event of an emergency, course schedules and assignments may be subject to change. If changes to graded activities are required, reasonable adjustments will be made, and you will be responsible for meeting revised deadlines.